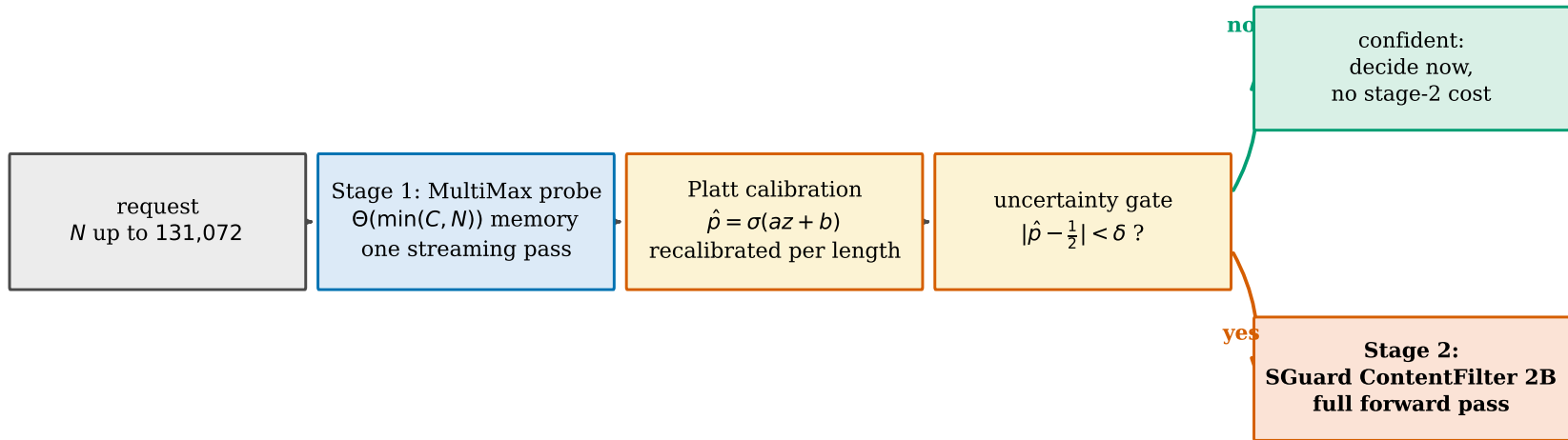


Two-stage safety cascade: a constant-memory gate in front of a heavyweight monitor



δ is the target-rate quantile of $|\hat{p} - \frac{1}{2}|$, so the escalation budget is set by the operator, not by a hard-coded constant.

Stage 2 runs on the escalated fraction only, so the compute saved is $(1 - \text{escalation rate})$ of the stage-2 cost.